Salesforce Certified JavaScript Developer

1. A team at Universal Containers works on a big project and uses yarn to deal with the project's dependencies.

A developer added a dependency to manipulate dates and pushed the updates to the remote repository. The rest of the team complains that the dependency does not get downloaded when they execute yarn.

What could be the reason for this?

- A. The developer added the dependency as a dev dependency, and YARN_ENV is set to production.
- B. The developer missed the option —save when adding the dependency.
- C. The developer added the dependency as a dev dependency, and NODE_ENV is set to production.
- D. The developer missed the option —add when adding the dependency.

---

2. Refer to the code below:

```javascript
let o = {
 get js() {
  let city1 = String('St. Louis');
  let city2 = String('New York');

  return {
   firstCity: city1.toLowerCase(),
   secondCity: city2.toLowerCase(),
  }
 }
}
```

```

What value can a developer expect when referencing `o.js.secondCity`?

- A. undefined

- B. An error

- C. 'New York'

- D. 'new york'

---

**3.** Refer to the code below:

```javascript
01 x = 3.14;
02
03 function myFunction() {
04   'use strict';
05   y = x;
06 }
07
08 z = x;
09 myFunction();
```

Considering the implications of `'use strict'` on line 04, which three statements describe the execution of the code?

- [ ] A. 'use strict' is hoisted, so it has an effect on all lines.

- [x] B. z is equal to 3.14.

- [ ] C. 'use strict' has an effect between line 04 and the end of the file.
```

- [x] D. 'use strict' has an effect only on line 05.

- [x] E. Line 05 throws an error.

---

4. Refer to the code below:

```javascript
01 let total = 10;
02 const interval = setInterval(() => {
03   total++;
04   clearInterval(interval);
05   total++;
06 }, 0);
07 total++;
08 console.log(total);
```

Considering that JavaScript is single-threaded, what is the output of line 08 after the code executes?

- A. 11

- B. 12

- C. 10

- D. 13

---

5. Refer to the code below:

```javascript
01 const objBook = {
```

```
title: 'JavaScript',
};
Object.preventExtensions(objBook);
const newObjBook = objBook;
newObjBook.author = 'Robert';
```

What are the values of `objBook` and `newObjBook` respectively?

- A. `{ title: "JavaScript" }`
  `{ title: "JavaScript" }`
- B. `{ author: "Robert", title: "JavaScript" }`
  `undefined`
- C. `{ author: "Robert" }`
  `{ author: "Robert", title: "JavaScript" }`
- D. `{ author: "Robert", title: "JavaScript" }`
  `{ author: "Robert", title: "JavaScript" }`

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6. A developer creates a simple webpage with an input field. When a user enters text in the input field and clicks the button, the actual value of the field must be displayed in the console.

Here is the HTML file content:

```html
<input type="text" value="Hello" name="input">
<button type="button">Display</button>
```

The developer wrote the JavaScript code below:

```javascript
01 const button = document.querySelector('button');
02 button.addEventListener('click', () => {
03   const input = document.querySelector('input');
04   console.log(input.getAttribute('value'));
05 });
```

When the user clicks the button, the output is always "Hello".

What needs to be done to make this code work as expected?

- A. Replace line 02 with `button.addCallback("click", function() {`

- B. Replace line 03 with `const input = document.getElementByIdName('input');`

- C. Replace line 04 with `console.log(input.value);`

- D. Replace line 02 with `button.addEventListener("onclick", function() {`

---

7. A developer at Universal Containers is creating their new landing page based on HTML, CSS, and JavaScript.

To ensure that visitors have a good experience, a script named `personalizeWebsiteContent` needs to be executed when the webpage is fully loaded (HTML content and all related files), in order to do some custom initializations.

Which implementation should be used to call `personalizeWebsiteContent` based on the business requirement above?

- A. Add a handler to the `personalizeWebsiteContent` script to handle the DOMContentLoaded event

- B. Add a listener to the `window` object to handle the `load` event

- C. Add a listener to the `window` object to handle the DOMContentLoaded event

- D. Add a handler to the `personalizeWebsiteContent` script to handle the `load` event

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9. A developer wants to create a simple image upload in the browser using the File API. The HTML is below:

```html
<input type="file" onchange="previewFile()">
<img src="" height="200" alt="Image preview..." />
```

The JavaScript portion is:

```javascript
01 function previewFile() {
02   const preview = document.querySelector('img');
03   const file = document.querySelector('input[type=file]').files[0];
04   // line 4 code
05   reader.addEventListener("load", () => {
06     preview.src = reader.result;
07   }, false);
08   //line 8 code
09 }
```

In lines 04 and 08, which code allows the user to select an image from their local computer, and to display the image in the browser?

- A. 04 `const reader = new File();`

  08 `if (file) reader.readAsDataURL(file);`

- B. 04 `const reader = new FileReader();`

  08 `if (file) reader.readAsDataURL(file);`

- C. 04 `const reader = new FileReader();`

08 `if (file) URL.createObjectURL(file);`

11. Refer to the code below:

```javascript
const exec = (item, delay) =>
  new Promise(resolve => setTimeout(() => resolve(item), delay));

async function runParallel() {
  const [result1, result2, result3] = await Promise.all(
    [exec('x', '100'), exec('y', '500'), exec('z', '100')]
  );
  return `parallel is done: ${result1}${result2}${result3}`;
}
```

Which two statements correctly execute the `runParallel()` function?

- A. `async runParallel().then(data);`
- B.

```javascript
runParallel().then(function(data){
  return data;
});
```

- C.

```javascript
runParallel().done(function(data){
  return data;
});
```

```

- D. `runParallel().then(data);`

---

**12.** A developer is leading the creation of a new browser application that will serve a single page application.

The team wants to use a new web framework Minimalist.js. The lead developer wants to advocate for a more seasoned web framework/library that already has a community around it.

Which two frameworks/libraries should the lead developer advocate for?

☑ A. React

☐ B. Koa

☒ C. Vue

☐ D. Express

---

**13.** Refer to the code below:

```javascript
let car1 = new Promise((_, reject) =>
  setTimeout(reject, 2000, "Car 1 crashed in"));
let car2 = new Promise(resolve => setTimeout(resolve, 1500, "Car 2 completed"));
let car3 = new Promise(resolve => setTimeout(resolve, 3000, "Car 3 completed"));

Promise.race([car1, car2, car3])
.then(value => {
  let result = '$(value) the race.';
})
.catch(err => {

```javascript
  console.log("Race is cancelled.", err);
});
```

What is the value of `result` when `Promise.race` executes?

- A. Car 2 completed the race.
- B. Car 3 completed the race.
- C. Race is cancelled.
- D. Car 1 crashed in the race.

14.

```javascript
function Tiger() {
  this.type = 'Cat';
  this.size = 'large';
}

let tony = new Tiger();
tony.roar = () => {
  console.log('They\'re great!');
};

function Lion() {
  this.type = 'Cat';
  this.size = 'large';
}

let leo = new Lion();
```

17 // Insert code here

18 leo.roar();

```

Which two statements could be inserted at line 17 to enable the function call on line 18?

- A. `leo.roar = () => { console.log('They\'re pretty good!'); };`

- B. `Object.assign(leo, tony);`

- C. `Object.assign(leo, Tiger);`

- D. `leo.prototype.roar = () => { console.log('They\'re pretty good!');`

---

**15.** A developer wants to use a `try...catch` statement to catch any error that `countSheep()` may throw and pass it to a `handleError()` function.

What is the correct implementation of the `try...catch`?

- A.

```javascript
try {
  setTimeout(function() {
    countSheep();
  }, 1000);
}
catch (e) {
  handleError(e);
}
```

- B.

```javascript
try {

  countSheep();
} finally {
  handleError(e);
}
```

- C.

```javascript
try {
  countSheep();
} handleError (e){
  catch(e);
}
```

- D.

```javascript
setTimeout(function() {
  try {
```

---

16. Refer to the code snippet:

```javascript
let array = [1, 2, 3, 4, 4, 5, 4, 4];
for (let i = 0; i < array.length; i++) {
  if (array[i] === 4) {
    array.splice(i, 1);
    i--;
  }
}

```

What is the value of `array` after the code executes?

- A. [1,2,3,4,5,4]

- B. [1,2,3,5]

- C. [1,2,3,4,5,4,4]

- D. [1,2,3,4,4,5,4]

---

17. Given the following code:

```javascript
let x = ('15' + 10) * 2;
```

What is the value of `x`?

- A. 1520

- B. 50

- C. 35

- D. 3020

---

18. A developer wants to set up a secure web server with Node.js. The developer creates a directory locally called `app-server`, and the first file is `app-server/index.js`.

Without using any third-party libraries, what should the developer add to `index.js` to create the secure web server?

- A. `const server = require('secure-server');`
```

- B. `const tls = require('tls');`

- C. `const http = require('http');`

- D. `const https = require('https');`

---

19. Refer to the code below:

```javascript
01 const event = new CustomEvent(
02   //Missing code
03 );
04 obj.dispatchEvent(event);
```

A developer needs to dispatch a custom event called `update` to send information about `recordId`.

Which two options could a developer insert at the placeholder in line 02 to achieve this?

- A. `{type : 'update', recordId : '123abc'}`

- B. `'update', { detail : { recordId : '123abc' } }`

- C. `'update', '123abc'`

- D. `'update', { recordId : '123abc' }`

21. Refer to the code below:

```javascript
01 let sayHello = () => {
02   console.log('Hello, World!');
03 };
```

Which code executes `sayHello` once, two minutes from now?

- A. `delay(sayHello, 120000);`
- B. `setTimeout(sayHello, 120000);`
- C. `setTimeout(sayHello(), 120000);`
- D. `setInterval(sayHello, 120000);`

23. Refer to the code below:

```javascript
let country = {
  get capital() {
    let city = Number("London");

    return {
      cityString: city.toString(),
    }
  }
}
```

Which value can a developer expect when referencing `country.capital.cityString`?

- [ ] A. An error
- [x] B. NaN
- [ ] C. undefined
- [ ] D. 'London'

---

24. Which statement accurately describes an aspect of promises?

- A. Arguments for the callback function passed to `.then()` are optional.

- B. `.then()` cannot be added after a catch.

- C. `.then()` manipulates and returns the original promise.

- D. In a `.then()` function, returning results is not necessary since callbacks will catch the result of a previous promise.

---

25. Given the code block below:

```javascript
01 function GameConsole(name) {
02   this.name = name;
03 }
04
05 GameConsole.prototype.load = function(gamename) {
06   console.log('${this.name} is loading a game: ${gamename}....');
07 }
08
09 function Console16bit(name) {
10   GameConsole.call(this, name);
11 }
12
13 Console16bit.prototype = Object.create(GameConsole.prototype);
14
15 // insert code here
16 console.log('${this.name} is loading a cartridge game: ${gamename}....');
17 }
```

```
18
19 const console16bit = new Console16bit('SNEGeneziz');
20 console16bit.load('Super Monic 3x Force');
```

What should a developer insert at line 15 to output the following message using the `load` method?

> SNEGeneziz is loading a cartridge game: Super Monic 3x Force...

- A. `Console16bit = Object.create(GameConsole.prototype).load = function(gamename) {`
- B. `Console16bit.prototype.load(gamename) = function() {`
- C. `Console16bit.prototype.load(gamename) {`
- D. `Console16bit.prototype.load = function(gamename) {`

---

26. Refer to the code:

```javascript
01 function Vehicle(name, price) {
02   this.name = name;
03   this.price = price;
04 }
05 Vehicle.prototype.priceInfo = function () {
06   return `Cost of the $(this.name) is $(this.price)$';
07 }
08 var ford = new Vehicle('Ford Fiesta', '20,000');
```

Given the requirement to refactor the code above to JavaScript class format, which class definition is correct?

- A.

```javascript
class Vehicle {
  constructor(name, price) {
    this.name = name;
    this.price = price;
  }
  priceInfo() {
    return 'Cost of the ${this.name} is ${this.price}$';
  }
}
```

- B.

```javascript
class Vehicle {
  vehicle(name, price) {
    this.name = name;
    this.price = price;
  }
  priceInfo() {
    return 'Cost of the ${this.name} is ${this.price}$';
  }
}
```

- C.

```javascript
class Vehicle {
  constructor() {
    this.name = name;
    this.price = price;
```

```
  }
  priceInfo() {
    return `Cost of the ${this.name} is ${this.price}$';
  }
}
```

- D.

```javascript
class Vehicle {
  constructor(name, price) {
    this.name = name;
    this.price = price;
  }
  priceInfo() {
    return `Cost of the ${this.name} is ${this.price}$';
  }
}
```

---

28. Given the code below:

```javascript
const copy = JSON.stringify([new String('false'), new Boolean(false), undefined]);
```

What is the value of `copy`?

- A. `'["false", false, null]'`

- B. `'[false, {}]'`

- C. `'["false", false, undefined]'`

- D. `'["false", {}]'`

---

29. Refer to the code below:

```javascript
let requestPromise = client.getRequest;

requestPromise().then((response) => {
  handleResponse(response);
});
```

A developer uses a client that makes a GET request that uses a Promise to handle the request. `getRequest` returns a Promise.

Which code modification can the developer make to gracefully handle an error?

- A.

```javascript
let requestPromise = client.getRequest;

try {
  requestPromise().then((response) => {
    handleResponse(response);
  });
} catch(error) {
  handleError(error);
}
```

```

- B.

```javascript
01 let requestPromise = client.getRequest;
02
03 try {
04   requestPromise().then((response) => {
05     handleResponse(response);
06   });
07 } catch(error) {
08   handleError(error);
09 }
```

- C.

```javascript
01 let requestPromise = client.getRequest;
02
03 requestPromise().then((response) => {
04   handleResponse(response);
05 }).catch((error) => {
06   handleError(error);
07 });
}
```

- D.

```javascript
01 let requestPromise = client.getRequest;
02
03 requestPromise().then((response) => {
04   handleResponse(response);
05 }).finally((error) => {

06   handleError(error);

07 });

```

---

30. A developer creates a new web server that uses Node.js. It imports a server library that uses events and callbacks for handling server functionality.

The server library is imported with `require` and is made available to the code by a variable named `server`. The developer wants to log any issues that the server has while booting up.

Given the code and the information the developer has, which code logs an error at boot time with an event?

- A.

```javascript
01 server.catch('error) => {
02   console.log('ERROR', error);
03 });
```

- B.

```javascript
01 server.error('error) => {
02   console.log('ERROR', error);
03 });
```

- C.

```javascript
01 server.on('error', (error) => {
02   console.log('ERROR', error);
03 });
```

- D.
```

```javascript
try {
  server.start();
} catch(error) {
  console.log('ERROR', error);
}
```

---

31. A developer wants to use a module called `DatePrettyPrint`. This module exports one default function called `printDate()`.

How can a developer import and use the `printDate()` function?

- A.

```javascript
import DatePrettyPrint() from '/path/DatePrettyPrint.js';
printDate();
```

- B.

```javascript
import DatePrettyPrint from '/path/DatePrettyPrint.js';
DatePrettyPrint.printDate();
```

- C.

```javascript
import printDate from '/path/DatePrettyPrint.js';
DatePrettyPrint.printDate();
```

- D.

```javascript
```

import printDate from '/path/DatePrettyPrint.js';

printDate();

```

---

**32.** Which three actions can be the code executes browser console?

- A. Run code that is not related to the page.

- B. View and change security cookies.

- C. Display a report showing the performance of a page.

- D. View, change, and debug the JavaScript code of the page.

- E. View and change the DOM of the page.

---

**33.** Which statement accurately describes the behavior of the `async/await` keywords?

- A. The associated function sometimes returns a promise.

- B. The associated function can only be called via asynchronous methods.

- C. The associated class contains some asynchronous functions.

- D. The associated function is asynchronous, but acts like synchronous code.

---

**34.** Refer to the code below:

```javascript
const pi = 3.1415926;
```
```

What is the data type of `pi`?

- [ ] A. Float
- [ ] B. Double
- [ ] C. Decimal
- [x] D. Number

---

35. Refer to the code below:

```javascript
function changeValue(obj) {
  obj.value = obj.value / 2;
}
const objA = {value: 10};
const objB = objA;

changeValue(objB);
const result = objA.value;
```

What is the value of `result` after the code executes?

- A. undefined
- B. 5
- C. NaN
- D. 10

---

36. A developer wants to create an object from a function in the browser using the code below.

```javascript
01 function Monster() { this.name = 'hello'; };
02 const m = Monster();
```

What happens due to the lack of the `new` keyword on line 02?

- A. The `m` variable is assigned the correct object.
- B. `window.name` is assigned to `'hello'` and the variable `m` remains undefined.
- C. `window.m` is assigned the correct object.
- D. The `m` variable is assigned the correct object but `this.name` remains undefined.

---

37. A class was written to represent items for purchase in an online store, and a second class representing items that are on sale at a discounted price. The constructor sets the name to the first value passed in. The pseudocode is below:

```javascript
class Item {
  constructor(name, price){
    ... // Constructor Implementation
  }
}

class SaleItem extends Item {
  constructor(name, price, discount){
    ... // Constructor Implementation
  }
}
```

```

There is a new requirement for a developer to implement a description method that will return a brief description for Item and SaleItem.

```javascript
let regItem = new Item('Scarf', 55);
let saleItem = new SaleItem('Shirt', 80, .1);
Item.prototype.description = function() { return 'This is a ' + this.name; }
console.log(regItem.description());
console.log(saleItem.description());

SaleItem.prototype.description = function() { return 'This is a discounted ' + this.name; }
console.log(regItem.description());
console.log(saleItem.description());
```

What is the output when executing the code above?

- A. This is a Scarf

  Uncaught TypeError: saleItem.description is not a function

  This is a Shirt

  This is a discounted Shirt

- B. This is a Scarf

  This is a Shirt

  This is a Scarf

  This is a discounted Shirt

- C. This is a Scarf

  Uncaught TypeError: saleItem.description is not a function
```

This is a Scarf

This is a discounted Shirt

- D. This is a Scarf

  This is a Shirt

  This is a discounted Scarf

  This is a discounted Shirt

---

38. Refer to the HTML below:

```html
<p> The current status of an Order: <span id="status"> In Progress </span> </p>
```

Which JavaScript statement changes the text 'In Progress' to 'Completed'?

- A. `document.getElementById(".status").innerHTML = 'Completed';`
- B. `document.getElementById("#status").innerHTML = 'Completed';`
- C. `document.getElementById("status").innerHTML = 'Completed';`
- D. `document.getElementById("status").Value = 'Completed';`

---

39. Universal Containers recently launched its new landing page to host a crowd-funding campaign. The page uses an external library to display some third-party ads. Once the page is fully loaded, it creates more than 50 new HTML items placed randomly inside the DOM, like the one in the code below:

```html
<!-- This is an ad -->
```

```
<div class="ad-library-item ad-hidden" onload="myFunction()">
  <img src="/ad-library/ad01.gif" />
</div>
```

All the elements include the same `ad-library-item` class. They are hidden by default, and they are randomly displayed while the user navigates through the page.

Tired of all the ads, what can the developer do to temporarily and quickly remove them?

- [ ] A. Use the browser console to execute a script that prevents the load event to be fired.
- [ ] B. Use the DOM inspector to prevent the load event to be fired.
- [x] C. Use the browser console to execute a script that removes all the elements containing the class `ad-library-item`.
- [ ] D. Use the DOM inspector to remove all the elements containing the class `ad-library-item`.

---

40. After user acceptance testing, the developer is asked to change the webpage background based on user's location. This change was implemented and deployed for testing.

The tester reports that the background is not changing, however it works as required when viewing on the developer's computer.

Which two actions will help determine accurate results?

☑ A. The tester should disable their browser cache.

☐ B. The developer should inspect their browser refresh settings.

☑ C. The tester should clear their browser cache.

☐ D. The developer should rework the code.

---

41. Refer to the code below:

```javascript
function execute() {
  return new Promise((resolve, reject) => reject());
}
let promise = execute();

promise
  .then(() => console.log('Resolved1'))
  .then(() => console.log('Resolved2'))
  .then(() => console.log('Resolved3'))
  .catch(() => console.log('Rejected'))
  .then(() => console.log('Resolved4'))
```

What is the result when the Promise in the `execute` function is rejected?

- A. Resolved1 Resolved2 Resolved3 Rejected Resolved4
- B. Rejected
- C. Resolved1 Resolved2 Resolved3 Resolved4
- D. Rejected Resolved4

---

42. The developer has a function that prints "Hello" to an input name. To test this, the developer created a function that returns "World". However, the following snippet does not print "Hello World".

```javascript
const sayHello = (name) => {
  console.log('Hello ', name);
};

```

05 const world = () => {

06   return 'World';

07 };

08

09 sayHello(world);

```

What can the developer do to change the code to print "Hello World"?

- A. Change line 2 to `console.log('Hello', name());`

- B. Change line 7 to `}();`

- C. Change line 9 to `sayHello(world)();`

- D. Change line 5 to `function world() {`

---

**43.** Refer to the string below:

```javascript

const str = 'Salesforce';

```

Which two statements result in the word 'Sales'?

- A. `str.substring(0, 5);`

- B. `str.substring(1, 5);`

- C. `str.substr(0, 5);`

- D. `str.substr(1, 5);`

```

44. A developer needs to test this function:

```javascript
const sum3 = (arr) => {
  if (!arr.length) return 0;
  if (arr.length === 1) return arr[0];
  if (arr.length === 2) return arr[0] + arr[1];
  return arr[0] + arr[1] + arr[2];
};
```

Which two assert statements are valid tests for this function?

- A. `console.assert(sum3([1, '2']) == 12);`
- B. `console.assert(sum3(['hello', 2, 3, 4]) === NaN);`
- C. `console.assert(sum3([-3, 2]) === -1);`
- D. `console.assert(sum3([0]) === 0);`

---

45. Refer to the code below:

```javascript
console.log('Start');
Promise.resolve('Success').then(function(value) {
  console.log('Success');
});
console.log('End');
```

What is the output after the code executes successfully?

- A. Start

  Success

  End

- B. Start

  End

  Success

- C. End

  Start

  Success

- D. Success

  Start

  End

---

46. A developer tries to retrieve all cookies, then sets a certain key value pair in the cookie. These statements are used:

```javascript
01 document.cookie;
02 document.cookie = 'key=John Smith';
```

What is the behavior?

- A. Cookies are read and the key value is set, the remaining cookies are unaffected.

- B. Cookies are read, but the key value is not set because the value is not URL encoded.

- C. Cookies are not read because line 01 should be `document.cookies`, but the key value is set and all cookies are wiped.

- D. Cookies are read and the key value is set, and all cookies are wiped.

---

47. A developer removes the HTML class attribute from the checkout button, so now it is simply:

```html
<button>Checkout</button>
```

There is a test to verify the existence of the checkout button, however it looks for a button with `class="blue"`. The test fails because no such button is found.

Which type of test category describes this test?

- [ ] A. False negative

- [ ] B. True positive

- [ ] C. True negative

- [ ] D. False positive

---

48. What are two unique features of functions defined with a fat arrow as compared to normal function definition?

- A. If the function has a single expression in the function body, the expression will be evaluated and implicitly returned.

- B. The function uses the `this` from the enclosing scope.

- C. The function receives an argument that is always in scope, called `parentThis`, which is the enclosing lexical scope.

- D. The function generates its own `this` making it useful for separating the function's scope from its enclosing scope.

---

49. A developer wants to leverage a module to print a price in pretty format, and has imported a method as shown below:

```javascript
import printPrice from '/path/PricePrettyPrint.js';
```

Based on the code, what must be true about the `printPrice` function of the `PricePrettyPrint` module for this import to work?

- [ ] A. `printPrice` must be a named export
- [ ] B. `printPrice` must be an all export
- [ ] C. `printPrice` must be the default export
- [ ] D. `printPrice` must be a multi export

---

50. Refer to the code below:

```javascript
let inArray = [ [1,2] , [3,4,5] ];
```

Which two statements result in the array `[ 1, 2, 3, 4, 5 ]`?

A. `[].concat(..inArray);`

B. `[].concat.apply(usArray, []);`

C. `[].concat::..usArray();`

D. `[].concat.apply({}, usArray);`

---

51. Refer to the code below:

```javascript
function myFunction(reassign) {
  let x = 1;
  var y = 1;

  if(reassign) {
    let x = 2;
    var y = 2;
    console.log(x);
    console.log(y);
  }

  console.log(x);
  console.log(y);
}
```

What is displayed when `myFunction(true)` is called?

- [ ] A. 2 2 2 2

- [ ] B. 2 2 1 2

- [ ] C. 2 2 1 1

- [ ] D. 2 2 undefined undefined

---

52. Given the following code:

```javascript
01 let x = null;
02 console.log(typeof x);
```

What is the output of line 02?

- A. "object"
- B. "undefined"
- C. "x"
- D. "null"

53. A developer creates a class that represents a news story based on the requirements that a Story should have a body, author, and view count. The code is shown below:

```javascript
01 class Story {
02   // Insert code here
03   this.body = body;
04   this.author = author;
05   this.viewCount = viewCount;
06 }
07 }
```

Which statement should be inserted in the placeholder on line 02 to allow for a variable to be set to a new instance of a Story with the three attributes correctly populated?

- A. `constructor() {`
- B. `super (body, author, viewCount) {`

- C. `function Story(body, author, viewCount) {`

- D. `constructor(body, author, viewCount) {`

---

54. Refer to the code below:

```javascript
01 const addBy = ?
02 const addByEight = addBy(8);
03 const sum = addByEight(50);
```

Which two functions can replace line 01 and return 58 to sum?

Choose 2 answers:

A.

```javascript
const addBy = function(num1){
  return function(num2){
    return num1 + num2;
  }
}
```

B.

```javascript
const addBy = function(num1){
  return num1 * num2;
}
```

C.

```javascript
```

const addBy = (num1) => num1 + num2;

```

D.

const addBy * function (num?) {

return function (num2) (

return numl + num2;```

---

**55.** Refer to the code below:

```javascript

let productSKU = '8675309';

```

A developer has a requirement to generate SKU numbers that are always 19 characters long, starting with 'sku', and padded with zeros.

Which statement assigns the value `sku0000000008675309`?

- A. `productSKU = productSKU.padEnd(16, '0').padStart('sku');`

- B. `productSKU = productSKU.padStart(16, '0').padStart(19, 'sku');`

- C. `productSKU = productSKU.padEnd(16, '0').padStart(19, 'sku');`

- D. `productSKU = productSKU.padStart(19, '0').padStart('sku');`

---

56. Given the JavaScript below:

```javascript

01 function filterDOM(searchString){

```
02   const parsedSearchString = searchString && searchString.toLowerCase();
03   document.querySelectorAll('.account').forEach(account => {
04   const accountName = account.innerHTML.toLowerCase();
05   account.style.display = accountName.includes(parsedSearchString) ? /* Insert code here */
06   });
07 }
```

Which code should replace the placeholder comment on line 05 to hide accounts that do not match the search string?

- A. `'block' : 'none'`
- B. `'none' : 'block'`
- C. `'hidden' : 'visible'`
- D. `'visible' : 'hidden'`

---

57. A developer is asked to fix some bugs reported by users. To do that, the developer adds a breakpoint for debugging.

```javascript
function Car(maxSpeed, color) {
  this.maxSpeed = maxSpeed;
  this.color = color;
}
let carSpeed = document.getElementById('carSpeed');
debugger;
let fourWheels = new Car(carSpeed.value, 'red');
```

When the code execution stops at the breakpoint on line 06, which two types of information are available in the browser console?

- A. A variable displaying the number of instances created for the Car object
- B. The information stored in the window.localStorage property
- C. The values of the carSpeed and fourWheels variables
- D. The style, event listeners and other attributes applied to the carSpeed DOM element

---

58. A developer has the function, shown below, that is called when a page loads.

```javascript
function onLoad() {
  console.log("Page has loaded!");
}
```

Where can the developer see the log statement after loading the page in the browser?

- A. On the browser JavaScript console
- B. On the terminal console running the web server
- C. In the browser performance tools log
- D. On the webpage console log

---

59. A developer needs to debug a Node.js web server because a runtime error keeps occurring at one of the endpoints.

The developer wants to test the endpoint on a local machine and make the request against a local server to look at the behavior. In the source code, the server.js file will start the server. The developer wants to debug the Node.js server only using the terminal.

Which command can the developer use to open the CLI debugger in their current terminal window?

A. node_start_inspect server.js`

B. node_inspect server.js`

C. node server.js --inspect`

D. `node -i server.js`

---

60. Refer to the following code block:

```javascript
let array = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11];
let output = 0;

for (let num of array) {
  if (output > 10) {
    break;
  }
  if (num % 2 == 0) {
    continue;
  }
  output += num;
}
```

What is the value of output after the code executes?

- A. 16

- B. 25

- C. 11

- D. 36

---

61. Refer to the code below:

```javascript
async function functionUnderTest(isOK) {
  if (isOK) return 'OK';
  throw new Error('not OK');
}
```

Which assertion accurately tests the above code?

- A. `console.assert(await functionUnderTest(true), 'OK')`
- B. `console.assert(await(functionUnderTest(true), 'not OK'))`
- C. `console.assert(functionUnderTest(true), 'OK')`
- D. `console.assert(await functionUnderTest(true), 'not OK')`

---

62. Which option is true about the strict mode in imported modules?

- A. Add the statement `use non-strict;` before any other statements in the module to enable not-strict mode.
- B. Imported modules are in strict mode whether you declare them as such or not.
- C. Add the statement `use strict = false;` before any other statements in the module to enable not-strict mode.
- D. A developer can only reference `notStrict()` functions from the imported module.

---

63. A developer is creating a simple webpage with a button. When a user clicks this button for the first time, a message is displayed.

The developer wrote the JavaScript code below, but something is missing. The message gets displayed every time a user clicks the button, instead of just the first time.

```javascript
01 function listen(event) {
02
03   alert('Hey! I am John Doe');
04
05 }
06 button.addEventListener('click', listen);
```

Which two code lines make this code work as required?

A. On line 04, use `event.stopPropagation();`

B. On line 02, use `event.first` to test if it is the first execution.

C. On line 06, add an option called `once` to `button.addEventListener()`.

D. On line 04, use `button.removeEventListener('click', listen);`

64. Choose 3 options.

Refer to the following code:

```javascript
let codeName = 'Bond';
let sampleText = 'The name is ${codeName}, Jim ${codeName}';
```

A developer is trying to determine if a certain substring is part of a string.

Which three code statements return true?

- A. `sampleText.includes('Jim');`

- B. `sampleText.includes('The', 1);`

- C. `sampleText.includes('Jim', 4);`

- D. `sampleText.indexOf('Bond') !== -1;`

- E. `sampleText.substring('Jim');`

65. Choose 1 option.

Refer to the code below:

```javascript
let car1 = new Promise((_, reject) =>
  setTimeout(reject, 2000, "Car 1 crashed in"));
let car2 = new Promise(resolve => setTimeout(resolve, 1500, "Car 2 completed"));
let car3 = new Promise(resolve => setTimeout(resolve, 3000, "Car 3 completed"));

Promise.race([car1, car2, car3])
.then(value => {
  let result = '${value} the race.';
})
.catch(err => {
  console.log("Race is cancelled.", err);
});
```

What is the value of result when `Promise.race` executes?

- A. Car 3 completed the race.

- B. Car 2 completed the race

- C. Race is cancelled.

- D. Car 1 crashed in the race.

66. Choose 1 option.

Given the following code:

```javascript
counter = 0;
const logCounter = () => {
```

03   console.log(counter);

04 };

05 logCounter();

06 setTimeout(logCounter, 2100);

07 setInterval(() => {

08   counter++;

09   logCounter();

10 }, 1000);

```

What will be the first four numbers logged?

- A. 0122

- B. 0123

- C. 0112

- D. 0012

67. Choose 2 options.

A developer has a fizzbuzz function that, when passed in a number, returns the following:

- 'fizz' if the number is divisible by 3.

- 'buzz' if the number is divisible by 5.

- 'fizzbuzz' if the number is divisible by both 3 and 5.

- empty string if the number is divisible by neither 3 or 5.

Which two test cases properly test scenarios for the fizzbuzz function?

- A. `let res = fizzbuzz(true); console.assert(res === '');`

- B. `let res = fizzbuzz(3); console.assert(res === '');`

- C. `let res = fizzbuzz(5); console.assert(res === 'fizz');`

- D. `let res = fizzbuzz(15); console.assert(res === 'fizzbuzz');`

68. Choose 1 option.

Refer to the following code:

```javascript

```
01 let obj = {
02   foo: 1,
03   bar: 2
04 }
05 let output = []
06
07 for (let something of obj) {
08   output.push(something);
09 }
10
11 console.log(output);
```

What is the value of output on line 11?

- A. An error will occur due to the incorrect usage of the for_of statement on line 07.

- B. [1, 2]

- C. ["foo", "bar"]

- D. ["foo:1", "bar:2"]

69. Choose 2 options.

Which two code snippets show working examples of a recursive function?

- A.

```javascript
const sumToTen = numVar => {
  if (numVar < 0)
    return;
  return sumToTen(numVar + 1);
};
```

- B.

```javascript
function factorial(numVar) {
```

  if (numVar < 0) return;

  if (numVar === 0) return 1;

  return numVar - 1;

}

```

- C.

```javascript
const factorial = numVar => {
  if (numVar < 0) return;
  if (numVar === 0) return 1;
  return numVar * factorial(numVar - 1);
};
```

- D.

```javascript
let countingDown = function(startNumber) {
  if (startNumber > 0) {
    console.log(startNumber);
    return countingDown(startNumber - 1);
  } else {
    Return startNumber);
  }
};
```

70. Choose 2 options.

Given two expressions var1 and var2, what are two valid ways to return the concatenation of the two expressions and ensure it is data type string?

- A. `String(var1).concat(var2)`

- B. `String.concat(var1 + var2)`

- C. `var1 + var2`
```

- D. `var1.toString() + var2.toString()`

71. Choose 1 option.

Refer to the code below:

```javascript
const myFunction = arr => {
  return arr.reduce((result, current) => {
    return result + current;
  }, 10);
}
```

What is the output of this function when called with an empty array?

- A. Returns 0

- B. Throws an error

- C. Returns NaN

- D. Returns 5

72. Choose 1 option.

Given the code below:

```javascript
function Person(name, email) {
  this.name = name;
  this.email = email;
}

const john = new Person('John', 'john@email.com');
const jane = new Person('Jane', 'jane@email.com');
const emily = new Person('Emily', 'emily@email.com');

let usersList = [john, jane, emily];
```

Which method can be used to provide a visual representation of the list of users and to allow sorting by the name or email attribute?

- A. `console.table(usersList);`

- B. `console.group(usersList);`

- C. `console.groupCollapsed(usersList);`

- D. `console.info(usersList);`

73. Choose 1 option.

Given the code below:

```javascript
setTimeout(() => {
  console.log(1);
}, 1100);
console.log(2);
new Promise((resolve, reject) => {
  setTimeout(() => {
    reject(console.log(3));
  }, 1000);
}).catch(() => {
  console.log(4);
});
console.log(5);
```

What is logged to the console?

- A. 25341

- B. 12435

- C. 12534

- D. 21435

74. Choose 1 option.

Refer to the code below:

```javascript
let timedFunction = () => {
  console.log('Timer called.');
};

let timerId = setInterval(timedFunction, 1000);
```

Which statement allows a developer to cancel the scheduled timed function?

- A. `clearInterval(timerId);`
- B. `removeInterval(timerId);`
- C. `removeInterval(timedFunction);`
- D. `clearInterval(timedFunction);`

75. Choose 1 option.

A developer at Universal Containers creates a new landing page based on HTML, CSS, and JavaScript.

To ensure that visitors have a good experience, a script named `personalizeWebsiteContent` needs to be executed to do some custom initialization when the webpage is fully loaded with HTML content and all related files.

Which statement should be used to call `personalizeWebsiteContent` based on the above business requirement?

- A. `document.addEventListener('DOMContentLoaded', personalizeWebsiteContent);`
- B. `document.addEventListener('onDOMContentLoaded', personalizeWebsiteContent);`
- C. `window.addEventListener('load', personalizeWebsiteContent);`
- D. `window.addEventListener('onload', personalizeWebsiteContent);`

76. Choose 1 option.

Refer to the following code:

```javascript
class Ship {
  constructor(size) {
    this.size = size;
  }
```

```
05 }
06
07 class FishingBoat extends Ship {
08   constructor(size, capacity){
09     //Missing code
10     this.capacity = capacity;
11   }
12   displayCapacity() {
13     console.log('The boat has a capacity of ${this.capacity} people.');
14   }
15 }
16
17 let myBoat = new FishingBoat('medium', 10);
18 myBoat.displayCapacity();
```

Which statement should be added to line 09 for the code to display 'The boat has a capacity of 10 people'?

- A. `super(size);`

- B. `ship.size = size;`

- C. `super.size = size;`

- D. `this.size = size;`

77. Choose 1 option.

A developer initiates a server with the file server.js and adds dependencies in the source code's package.json that are required to run the server.

Which command should the developer run to start the server locally?

- A. `node start`

- B. `npm start`

- C. `npm start server.js`

- D. `start server.js`

78. Choose 3 options.

Given the code below:

```javascript
let numValue = 1982;
```

Which three code segments result in a correct conversion from number to string?

- A. `let strValue = numValue.toText();`
- B. `let strValue = String(numValue);`
- C. `let strValue = '' + numValue;`
- D. `let strValue = numValue.toString();`
- E. `let strValue = (String)numValue;`

79. Choose 2 options.

Which two console logs output NaN?

- A. `console.log(10 / 0);`
- B. `console.log(parseInt('two'));`
- C. `console.log(10 / Number('5'));`
- D. `console.log(10 / 'five');`

80. Choose 2 options.

A developer uses a parsed JSON string to work with user information as in the block below:

```javascript
01 const userInformation = {
02   "id": "user-01",
03   "email": "user01@universalcontainers.demo",
04   "age": 25
05 };
```

Which two options access the email attribute in the object?

- A. `userInformation.email`

- B. `userInformation.get("email")`

- C. `userInformation["email"]`

- D. `userInformation[email]`

81. Given a value, which three options can a developer use to detect if the value is NaN?

Choose 3 answers
A. value === Number.NaN
B. value == NaN
C. Object.is(value, NaN)
D. value !== value
E. Number.isNaN(value)

82. Choose 3 options.

Which three browser specific APIs are available for developers to persist data between page loads?

- A. localStorage

- B. indexedDB

- C. cookies

- D. global variables

- E. IIFEs

83. Choose 2 options.

Given a value, which two options can a developer use to detect if the value is NaN?

- A. `value === Number.NaN`

- B. `value == NaN`

- C. `isNaN(value)`

- D. `Object.is(value, NaN)`

84. Choose 1 option.

Refer to the code below:

```javascript
```

```
01 new Promise((resolve, reject) => {
02   const fraction = Math.random();
03   if (fraction > 0.5) reject('fraction > 0.5, ' + fraction);
04   resolve(fraction);
05 })
06 .then(() => console.log('resolved'))
07 .catch((error) => console.error(error))
08 .finally(() => console.log('when am I called?'));
```

When does `Promise.finally` on line 08 get called?

- A. When rejected
- B. When resolved and settled
- C. When resolved
- D. When resolved or rejected

86. Choose 1 option.

Refer to the HTML below:

```html
<div id="main">
  <ul>
    <li>Leo</li>
    <li>Tony</li>
    <li>Tiger</li>
  </ul>
</div>
```

Which JavaScript statement results in changing "Leo" to "The Lion"?

- A. `document.querySelectorAll('#main #Leo').innerHTML = 'The Lion';`
- B. `document.querySelector('#main li:second-child').innerHTML = 'The Lion';`
- C. `document.querySelectorAll('#main li,Leo').innerHTML = 'The Lion';`

- D. `document.querySelector('#main li:nth-child(1)').innerHTML = 'The Lion';`

87. Choose 1 option.

Refer to the following code block:

```javascript
class Animal {
  constructor(name) {
    this.name = name;
  }

  makeSound() {
    console.log('${this.name} is making a sound.');
  }
}

class Dog extends Animal {
  constructor(name) {
    super(name);
    this.name = name;
  }
  makeSound() {
    console.log('${this.name} is barking.');
  }
}

let myDog = new Dog('Puppy');
myDog.makeSound();
```

What is the console output?

- A. > Uncaught ReferenceError

- B. > Undefined

- C. > Puppy is barking.

88. Choose 1 option.

Refer to the code below:

```javascript
const pi = 3.1415926;
```

What is the data type of pi?

- A. Number

- B. Float

- C. Double

- D. Decimal

89. Choose 1 option.

Refer to the code snippet:

```javascript
01 function getAvailableilityMessage(item) {
02   if (getAvailableility(item)) {
03     var msg = "Username available";
04   }
05   return msg;
06 }
```

A developer writes this code to return a message to a user attempting to register a new username. If the username is available, a variable named msg is declared and assigned a value on line 03.

What is the return value of msg when `getAvailableilityMessage("newUserName")` is executed and `getAvailableility("newUserName")` returns false?

- A. "newUserName"

- B. "msg is not defined"

- C. undefined

- D. "Username available"

90. Choose 1 option.

Refer to the code below:

```javascript
function changeValue(param) {
  param = 5;
}
let a = 10;
let b = a;

changeValue(b);
const result = a + ' - ' + b;
```

What is the value of result when the code executes?

- A. 5-5

- B. 5-10

- C. 10-5

- D. 10-10

91. Choose 1 option.

Universal Containers (UC) just launched a new landing page, but users complain that the website is slow. A developer found some functions that might cause this problem. To verify this, the developer decides to execute everything and log the time each of these three suspicious functions consumes.

```javascript
console.time('Performance');

maybeAHeavyFunction();
```

```
thisCouldTakeTooLong();

orMaybeThisOne();

console.endTime('Performance');
```

Which function can the developer use to obtain the time spent by every one of the three functions?

- A. `console.timeLog()`
- B. `console.trace()`
- C. `console.timeStamp()`
- D. `console.getTime()`

92. Choose 1 option.

Given the code below:

```javascript
const delay = async delay => {
  return new Promise((resolve, reject) => {
    console.log(1);
    setTimeout(resolve, delay);
  });
};

const callDelay = async () => {
  console.log(2);
  const yup = await delay(1000);
  console.log(3);
};

console.log(4);
callDelay();
```

16 console.log(5);

```

What is logged to the console?

- A. 4 2 1 5 3

- B. 4 2 1 5 3

- C. 1 4 2 3 5

- D. 4 5 1 2 3

93. Choose 1 option.

A developer is setting up a Node.js server and is creating a script at the root of the source code, index.js, that will start the server when executed. The developer declares a variable that needs the folder location that the code executes from.

Which global variable can be used in the script?

- A. `__filename`

- B. `window.location`

- C. `this.path`

- D. `__dirname`

94. Choose 1 option.

Refer to the code below:

```javascript
flag();

function flag() {

  console.log('flag');

}


const anotherFlag = () => {

  console.log('another flag');

}

```javascript
anotherFlag();
```

What is result of the code block?

- A. The console logs only 'flag'.

- B. An error is thrown.

- C. The console logs 'flag' and then an error is thrown.

- D. The console logs 'flag' and 'another flag'.

95. Choose 1 option.

Refer to the code below:

```javascript
let first = 'Who';
let second = 'What';
try {
  try {
    throw new Error('Sad trombone');
  } catch (err) {
    first = 'Why';
    throw err;
  } finally {
    second = 'When';
  }
} catch (err) {
  second = 'Where';
}
```

What are the values for first and second once the code executes?

- A. first is Who and second is Where.

- B. first is Why and second is Where.

- C. first is Who and second is When.

- D. first is Why and second is When.

96. Choose 1 option.

Refer to the code below:

```javascript
const searchText = 'Yay! Salesforce is amazing!';

let result1 = searchText.search(/sales/i);
let result2 = searchText.search(/sales/);

console.log(result1);
console.log(result2);
```

After running this code, which result is displayed on the console?

- A. > 5

  > undefined

- B. > 5

  > 0

- C. > true

  > false

- D. > 5

  > -1

97. Choose 2 options.

A developer is leading the creation of a new web server for their team that will fulfill API requests from an existing client.

The team wants a web server that runs on Node.js, and they want to use the new web framework Minimalist.js. The lead developer wants to advocate for a more seasoned back-end framework that already has a community around it.

Which two frameworks could the lead developer advocate for?

-A. Angular

- B. Next

- C. Gatsby

- D. Next.js

98. Choose 1 option.

A developer wrote the following code:

```javascript
01 let x = object.value;
02
03 try {
04   handleObjectValue(x);
05 } catch(error) {
06   handleError(error);
07 }
```

The developer has a `getNextValue` function to execute after `handleObjectValue()`, but does not want to execute `getNextValue()` if an error occurs.

How can the developer change the code to ensure this behavior?

- A.

```javascript
03 try {
04   handleObjectValue(x);
05 } catch(error) {
06   handleError(error);
07 } then {
08   getNextValue();
09 }
```

- B.

```javascript
try {
  handleObjectValue(x);
  getNextValue();
} catch(error) {
  handleError(error);
}
```

- C.

```javascript
try {
  handleObjectValue(x);
} catch(error) {
  handleError(error);
}
getNextValue();
```

- D.

```javascript
try {
  handleObjectValue(x);
} catch(error) {
  handleError(error);
} finally {
  getNextValue();
}
```

99. Choose 1 option.

Considering type coercion, what does the following expression evaluate to?

```javascript
```

true + '13' + NaN
```

- A. 'true13NaN'
- B. '113NaN'
- C. 14
- D. 'true13'

100. Choose 1 option.

Refer to the code below:

```javascript
01 const server = require('server');
02
03 // Insert code here
```

A developer imports a library that creates a web server. The imported library uses events and callbacks to start the server.

Which code should be inserted at line 03 to set up an event and start the web server?

- A. `server.on('connect', (port) => { console.log('Listening on', port); });`
- B. `server.start();`
- C. `server((port) => { console.log('Listening on', port); });`
- D. `server();`

101. Choose 1 option.

At Universal Containers, every team has its own way of copying JavaScript objects. The code snippet shows an implementation from one team:

```javascript
function Person() {
  this.firstName = "John";
  this.lastName = "Doe";
  this.name = () => {
    console.log('Hello ${this.firstName} ${this.lastName}');
```
```

```
06   }

07 }

08

09 const john = new Person();

10 const dan = JSON.stringify(JSON.parse(john));

11 dan.firstName = 'Dan';

12 dan.name();
```

What is the output of the code execution?

- A. Hello John Doe
- B. Hello Dan Doe
- C. TypeError: dan.name is not a function
- D. Hello Dan

102. Choose 1 option.

Given the HTML below:

```html
<div>
  <div id="row-uc">Universal Containers</div>
  <div id="row-as">Applied Shipping</div>
  <div id="row-bt">Burlington Textiles</div>
</div>
```

Which statement adds the `priority-account` CSS class to the Applied Shipping row?

- A. `document.querySelectorAll('#row-as').classList.add('priority-account');`
- B. `document.querySelector('#row-as').classList.add('priority-account');`
- C. `document.querySelector('#row-as').classes.push('priority-account');`
- D. `document.getElementById('row-as').addClass('priority-account');`

103. Choose 1 option.

A developer is required to write a function that calculates the sum of elements in an array but is getting undefined every time the code is executed. The developer needs to find what is missing in the code below.

```javascript
01 const sumFunction = arr => {
02   return arr.reduce((result, current) => {
03     //
04     result += current;
05     //
06   }, 10);
07 };
```

Which line replacement makes the code work as expected?

- A. `03 if(arr.length == 0) { return 0; }`

- B. `04 result = result + current;`

- C. `02 arr.map((result, current) => {`

- D. `05 return result;`

104. Choose 2 options.

A developer copied a JavaScript object:

```javascript
01 function Person() {
02   this.firstName = "John";
03   this.lastName = "Doe";
04   this.name = () => '${this.firstName},${this.lastName}';
05 }
06
07 const john = new Person();
08 const dan = Object.assign(john);
09 dan.firstName = 'Dan';
```

```

How does the developer access dan's firstName, lastName?

- A. `dan.firstName() + dan.lastName()`

- B. `dan.name()`

- C. `dan.name`

- D. `dan.firstName + dan.lastName`

105. Choose 1 option.

A developer publishes a new version of a package with bug fixes but no breaking changes. The old version number was `2.1.1`.

What should the new package version number be based on semantic versioning?

- A. `2.1.2`

- B. `2.2.0`

- C. `2.2.1`

- D. `3.1.1`

106. Choose 1 option.

A developer wants to use a module named `universalContainerslib` and then call functions from it.

How should a developer import every function from the module and then call the functions `foo` and `bar`?

- A.

```javascript

import * as lib from '/path/universalContainerslib.js';

lib.foo();

lib.bar();

```

- B.

```javascript

import * from '/path/universalContainerslib.js';

universalContainerslib.foo();

universalContainerslib.bar();

```

- C.

```javascript
import all from '/path/universalContainerslib.js';
universalContainerslib.foo();
universalContainerslib.bar();
```

- D.

```javascript
import {foo, bar} from '/path/universalContainerslib.js';
foo();
bar();
```

107. Choose 1 option.

Refer to the following object:

```javascript
const cat = {
  firstName: 'Fancy',
  lastName: 'Whiskers',
  get fullName(){
    return this.firstName + ' ' + this.lastName;
  }
};
```

How can a developer access the fullName property for cat?

- A. `cat.fullName()`
- B. `cat.get.fullName`
- C. `cat.function.fullName()`
- D. `cat.fullName`
```

108. Choose 1 option.

A developer has two ways to write a function:

Option A:

```javascript
function Monster() {
  this.growl = () => {
    console.log("Grr!");
  }
}
```

Option B:

```javascript
function Monster() {};
Monster.prototype.growl = () => {
  console.log("Grr!");
}
```

After deciding on an option, the developer creates 1000 monster objects.

How many growl methods are created with Option A and Option B?

- A. 1000 growl methods are created regardless of which option is used.

- B. 1 growl method is created regardless of which option is used.

- C. 1000 growl methods are created for Option A. 1 growl method is created for Option B.

- D. 1 growl method is created for Option A. 1000 growl methods are created for Option B.

109. Choose 1 option.

Refer to the code below:

```html
01 <html lang="en">
02 <table onclick="console.log('Table log');">
03   <tr id="row1">
04     <td>Click me!</td>
05   </tr>
06 </table>
07 <script>
08   function printMessage(event) {
09     console.log('Row log');
10     event.stopPropagation();
11   }
12
13   let elem = document.getElementById('row1');
14   elem.addEventListener('click', printMessage, false);
15 </script>
16 </html>
```

Which code change should be done for the console to log the following when 'Click me!' is clicked?

> Row log

> Table log

- A. Change line 14 to `elem.addEventListener('click', printMessage, true);`
- B. Remove lines 13 and 14
- C. Change line 10 to `event.stopPropagation(false);`
- D. Remove line 10

110. Choose 3 options.

```javascript
function Animal(size, type) {
```

```
  this.type = type || 'Animal';
  this.canTalk = false;
}

Animal.prototype.speak = function() {
  if (this.canTalk) {
    console.log("It spoke!");
  }
};

let Pet = function(size, type, name, owner) {
  Animal.call(this, size, type);
  this.size = size;
  this.name = name;
  this.owner = owner;
}

Pet.prototype = Object.create(Animal.prototype);
let pet1 = new Pet();
```

Given the code above, which three properties are set for pet1?

- A. speak
- B. owner
- C. canTalk
- D. name
- E. type

111. Choose 2 options.

A developer wrote the following code to test a sum3 function that takes in an array of numbers and returns the sum of the first three numbers in the array. The test passes:

```javascript
01 let res = sum3([1, 2, 3]);
02 console.assert(res === 6);
03
04 res = sum3([1, 2, 3, 4]);
05 console.assert(res === 6);
```

A different developer made changes to the behavior of sum3 to instead sum all of the numbers present in the array.

Which two results occur when running the test on the updated sum3 function?

- A. The line 02 assertion fails.

- B. The line 05 assertion passes.

- C. The line 05 assertion fails.

- D. The line 02 assertion passes.

112. Choose 1 option.

Given the code below:

```javascript
01 function Person() {
02   this.firstName = 'John';
03 }
04
05 Person.proto = {
06   job: x => 'Developer'
07 });
08
09 const myFather = new Person();
10 const result = myFather.firstName + ' ' + myFather.job();
```

What is the value of result when line 10 executes?

- A. Error: myFather.job is not a function

- B. undefined Developer

- C. John Developer

- D. John undefined

113. Choose 1 option.

Given the code below:

```javascript
01 setCurrentUrl();
02 console.log("The current URL is: " + url);
03
04 function setCurrentUrl() {
05   url = window.location.href;
06 }
```

What happens when the code executes?

- A. The url variable has global scope and line 02 throws an error.

- B. The url variable has global scope and line 02 executes correctly.

- C. The url variable has local scope and line 02 executes correctly.

- D. The url variable has local scope and line 02 throws an error.

114. **Given the JavaScript below:**

```javascript
function filterDOM(searchString){
  const parsedSearchString = searchString && searchString.toLowerCase();
  document.querySelectorAll('.account').forEach(account => {
    const accountName = account.innerHTML.toLowerCase();
    account.style.display = accountName.includes(parsedSearchString) ? /* Insert code here */;
  });
```

}
```

Which code should replace the placeholder comment on line 05 to hide accounts that do not match the search string?

- A. `'block' : 'none'`

- B. `'hidden' : 'visible'`

- C. `'visible' : 'hidden'`

- D. `'none' : 'block'`

115. A developer removes the HTML class attribute from the checkout button, so now it is simply:

`&lt;button&gt;Checkout&lt;/button&gt;`

There is a test to verify the existence of the checkout button, however it looks for a button with `class="blue"`. The test fails because no such button is found.

Which type of test category describes this test?

- A. True negative

- B. True positive

- C. False negative

- D. False positive

116. **Refer to the code snippet:**

```javascript
01 let array = [1, 2, 3, 4, 4, 5, 4, 4];
02 for (let i = 0; i < array.length; i++) {
03   if (array[i] === 4) {
04     array.splice(i, 1);
05     i--;
06   }
```

What is the value of array after the code executes?
```

- A. `[1, 2, 3, 4, 5, 4, 4]`
- B. `[1, 2, 3, 4, 4, 5, 4]`
- C. `[1, 2, 3, 4, 5, 4]`
- D. `[1, 2, 3, 5]`

117. A team at Universal Containers works on a big project and use yarn to deal with the project's dependencies.

A developer added a dependency to manipulate dates and pushed the updates to the remote repository. The rest of the team complains that the dependency does not get downloaded when they execute `yarn`.

What could be the reason for this?

- A. The developer missed the option `--add` when adding the dependency.
- B. The developer added the dependency as a dev dependency, and `NODE_ENV` is set to production.
- C. The developer added the dependency as a dev dependency, and `YARN_ENV` is set to production.
- D. The developer missed the option `--save` when adding the dependency.

118. A developer creates a new web server that uses Node.js. It imports a server library that uses events and callbacks for handling server functionality.

The server library is imported with `require` and is made available to the code by a variable named `server`. The developer wants to log any issues that the server has while booting up.

Given the code and the information the developer has, which code logs an error at boot time with an event?

- A.

```javascript
server.error((error) => {
  console.log('ERROR', error);
});
```

- B.

```javascript
```

```
server.catch((error) => {
  console.log('ERROR', error);
});
```

- C.

```javascript
try {
  server.start();
} catch(error) {
  console.log('ERROR', error);
}
```

- D.

```javascript
server.on('error', (error) => {
  console.log('ERROR', error);
});
```

119. A developer wants to use a `try...catch` statement to catch any error that `countSheep()` may throw and pass it to a `handleError()` function.

What is the correct implementation of the `try...catch`?

- A.

```javascript
setTimeout(function() {
  try {
    countSheep();
  } catch (e) {
    handleError(e);
```

  }
 }, 1000);
 ```

- B.
 ```javascript
 try {
  countSheep();
 } finally {
  handleError(e);
 }
 ```

- C.
 ```javascript
 try {
  countSheep();
 } handleError (e){
  catch(e);
 }
 ```

- D.
 ```javascript
 try {
  setTimeout(function() {
   countSheep();
  }, 1000);
 } catch (e) {
  handleError(e);
 }
 ```
 ```

120. A class was written to represent items for purchase in an online store, and a second class representing items that are on sale at a discounted price. The constructor sets the name to the first value passed in. The pseudocode is below:

```javascript
class Item {
  constructor(name, price){
    ... // Constructor Implementation
  }
}

class SaleItem extends Item {
  constructor(name, price, discount){
    ... // Constructor Implementation
  }
}
```

There is a new requirement for a developer to implement a description method that will return a brief description for Item and SaleItem.

```javascript
let regItem = new Item('Scarf', 55);
let saleItem = new SaleItem('Shirt', 80, .1);
Item.prototype.description = function() { return 'This is a ' + this.name; }
console.log(regItem.description());
console.log(saleItem.description());

SaleItem.prototype.description = function() { return 'This is a discounted ' + this.name; }
```

What is the output when executing the code above?

- A.
  ```
  This is a Scarf
  ```

Uncaught TypeError: saleItem.description is not a function

This is a Scarf

This is a discounted Shirt

```

- B.

```

This is a Scarf

This is a Shirt

This is a Scarf

This is a discounted Shirt

```

- C.

```

This is a Scarf

This is a Shirt

This is a discounted Scarf

This is a discounted Shirt

```

- D.

```

This is a Scarf

Uncaught TypeError: saleItem.description is not a function

This is a Shirt

This is a discounted Shirt

```

121. **Given the following code:**

`let x = ('15' + 10) * 2;`

What is the value of x?
```

- A. `1520`

- B. `3020`

- C. `50`

- D. `35`

122. A developer has an isDeg function that takes one argument, `pts`. They want to schedule the function to run every minute.

What is the correct syntax for scheduling this function?

Options:

A. setInterval (isDeg('cat'), 60000);`

B. setInterval(isDeg, 60000, 'cat'));`

C. setTimeout(isDeg, 60000, 'cat'));`

D. setTimeout(isDeg('cat'), 60000);`

123. Refer to the code below:

```javascript
let cat3 = new Promise(resolve => #witnessval(resolve, 3000, "Cat 3 completes");

Promise.exec([cat1, cat2, cat3])
.then(value => {
let result = "${value} the race.");
}
.catch(str => {
console.log("Race is cancelled:", err);
});
```

What is the value of `result` when `Promise.race` executed?

Options:

A. Car 2 completed the race.

B. Car 1 crashed on the race.

C. Race is cancelled.

D. Car 3 completed the race.

124. Refer to the code below:

```javascript
function changeValue(obj) {
  obj.value = obj.value / 2;
}
const objA = {value: 10};
const objB = objA;

changeValue(objB);
const result = objA.value;
```

What is the value of result after the code executes?

Options:

A. low

B. 10

C. 5

D. undefined

125. Refer to the code below:

```javascript
for (1st number = 2; number <= 5; number += 1) {
  // faster code statement here
}
```

The developer needs to insert a code statement in the location shown. The code statement has these requirements:

1. Does not require an import

2. Logs an error when the Boolean statement evaluates to false

3. Works in both the browser and Node.js

Which statement meets these requirements?

Options:

A. `console.classy(number + 2 === 0);`

B. `assert(number + 2 === 0);`

C. `console.assert(number + 2 === 0);`

D. `console.error(number + 2 === 0);`

127. Refer to the string below:

`const str = 'Salesforce';`

Which two statements result in the word "Sales"?

Options:

A. `str.substring(0, 5);`

B. `str.substr(s, 5);`

C. `str.substring(0, 5);`

D. `str.substr(0, 5);`

128. : try...catch Implementation**

A developer wants to use a `try...catch` statement to catch any error that `countsDeep()` may show and pass it to `handleError()` function.

What is the correct implementation of the `try...catch`?

Options:

A.

```javascript
try {
  countsDeep();
} handleError (e){
  catch(e);
}
```

B.

```javascript
setTimeout(function() {
  try {
    countsDeep();
  } catch (e) {
    handleError(e);
  }
}, 1000);
```

```

C.

```javascript
try {
  setTimeout(function() {
    countsDeep();
  }, 1000);
} catch (e) {
  handleError(e);
}

D.

```javascript
try {
  setTimeout(function() {
    countSheep();
  }, 1000);
} catch (e) {
  handleError(e);
}



129. Which statement parses successfully?

**Options:**

A. `JSON.parse("foo");`

B. `JSON.parse("foo");`

C. `JSON.parse("foo");`

D. `JSON.parse("foo");`
```

130. Which two actions can be done using the JavaScript browser console?

Choose 3 answers

A. Run code that’s not related to the page.

B. View and change the DOM of the page.

C. Display a report showing the performance of a page.

D. Change the DOM and the JavaScript code of the page.

E. View and change security cookies .

131. Given the code below:

```javascript
static delay = async delay => {
  return new Promise(resolve => {
    setTimeout(resolve, delay);
  });
};
static asyncCall = async () => {
  await delay(1000);
  console.log(1);
};
console.log(2);
asyncCall();
console.log(3);
```

What is logged to the console?

A. 1 2 3

B. 1 3 2

C. 2 1 3

D. 2 3 1

132. Refer to the following code:

```javascript
let page2stream = "The quick brown fox jumps";
```

A developer needs to determine if a certain substring is part of a string.

Which three expressions return true for the given substring?

Choose 3 answers

A. sampleText.includes('fox');

B. sampleText.indexOf('quick' > -4);

C. sampleText.indexOf(fox);

D. sampleText.include('fox') !== -1;

E. . sampleText.indexOf('Quick') !== -1;

133. Given two expressions, `exp1` and `exp2`, which two valid ways to return the logical AND of the two expressions and ensure it is a boolean?

Choose 2 answers

A. Boolean (var1 66 var2)

B. `Boolean(exp1) && Boolean(exp2)`

C. `exp1 & exp2`

D. ` Boolean(var1) 66 Boolean(var2)`

134. A developer publishes a new version of a package with new features that do not break backward compatibility. The previous version number was 1.1.3.

Following semantic versioning formats, what should the new package version number be?

A. 1.2.3

B. 1.1.4

C. 2.0.0

D. 1.2.0

135. Given the code below:

```javascript
01 function myFunction() {
02   a = a + b;
03   var b = 1;
04 }
05 myFunction();
06 console.log(a);
07 console.log(b);
```

What is the expected output?

A. Line 02 throws a reference error, therefore line 03 is never executed.

B. Both line 02 and 03 are executed, but the values printed are undefined.

C. Both line 02 and 03 are executed, and the variables are hoisted.

D. Line 08 output the variable, but line 09 throws an error.

136. Refer to the code below:

```javascript
(function first() {
  var a = 'Hello';
  function second() {
    console.log(a);
  }
  return second;
})();
```

Why does the function second have access to variable a?

A. Inner function's scope

B. Outer function's scope

C. Prototype Chain

D. Hoisting

137. Which statement allows a developer to update the browser navigation history without a page refresh?

A. `window.customHistory.pushState(newStateObject, '', null);`

B. `window.history.createState(newStateObject, '');`

C. `window.history.pushState(newStateObject, '', null);`

D. `window.history.updateState(newStateObject, '');`

138. A developer creates a generic function to log custom messages in the console. To do this, the function below is implemented.

```javascript
01 function logStatus(status) {
02   console./*answer goes here*/ ("The status is %s", status);
03 }
```

Which three console logging methods allow the use of string substitution in line 02?

Choose 3 answers

A. `message`

B. `log`

C. `assert`

D. `info`

E. `error`

139. Refer to the code below:

```html
<html>
  <body>
    <div id="logo">Hello Logo!</div>
    <button id="test">Click me</button>
```

```
  </body>
  <script>
    function printMessage() {
      console.log('This is a test message');
    }
    let el = document.getElementById('test');
    el.addEventListener("click", printMessage, false);
  </script>
</html>
```

Which code change should be made for the console to log only once when the button is clicked?

A. Add event. removefventListener(); 10 window. onLoad event handier

B. Add event. removetventiistener(); tO printlessage function

C. Add event. stopPropagation(); 10 printivessage function

D. Add event.stoppropagation(); to winontdoado evewnt handier

140. Refer to the following code:

```html
<html lang="en">
<body>
  <button class="secondary">Save draft</button>
  <button class="primary">Save and close</button>
</body>
<script>
  function displaySaveMessage(event) {
    console.log('Save message.');
  }
  function displaySuccessMessage(event) {
    console.log('Success message.');
  }
```

```
  window.onload = function() {
    document.querySelector('.secondary')
      .addEventListener('click', displaySaveMessage, true);
    document.querySelector('.primary')
      .addEventListener('click', displaySuccessMessage, true);
  }
</script>
</html>
```

What will the console show when the button is clicked?

A. >Outer message

B. >Outer message

   >Inner message

C. >Inner message

D. >Inner message

   >Outer message

141. Refer to the code below:

```javascript
01 let a = "*";
02 let b = "**";
03 / 7 x = 3;
04 console.log(a);
```

What is displayed when the code executes?

A. ReferenceError: a is not defined

B. a

C. undefined

D. null

142. Refer to the following code that imports a module named `utils`:

```javascript
import {foo, bar} from "./utils.js";

foo();

bar();
```

Which two implementations of `utils.js` support `foo` and `bar` such that the code above runs without error?

Choose 2 answers

A.

```javascript
const foo = () => { return 'foo'; }

const bar = () => { return 'bar'; }

export {foo, bar};
```

B.

```javascript
const foo = () => { return 'foo'; }

const bar = () => { return 'bar'; }

export default {foo, bar};
```

C.

```javascript
// utils.js

import {foo} from "./helpers/utils.js";

export {foo, bar};
```

D.

```javascript
export default class {

 foo() { return 'foo'; }
```

bar() { return 'bar'; }

}

143. A developer has an irresponsible module that contains multiple functions.

What kind of export should be leveraged so that multiple functions can be used?

A. `multi`

B. `default`

C. ` all`

D. named

144. Refer to the following code:

```javascript
01 let obj = {
02 fool 1,
03 bari 2
04 )
05 let output = [7#
08 output .push (something);
09 )
10
11 conssie.tog (output):
```

What is the outpouf tlin e 117

A. [bar, *foo")

B. 11,2)

C. [foo", “bar"]

D. ["foo:1", "bar:2"]

145. The developer wants to test the array shown:

```javascript
const arr = Array(5).fill(0);
```

Which two tests are the most accurate for this array?

Choose 2 answers

A. `console.assert(arr.length === 0);`

B. `console.assert(arr[5] === 0 && arr.length === 0);`

C. ` arr.forEach(eien => console.asert(elen ===0);

D. `console.assert(arr.length === 5);`

146. Refer to the following object.

```javascript
const dog = {
  firstName: 'Beau',
  lastName: 'Boo',
  get fullName() {
    return this.firstName + ' ' + this.lastName;
  }
};
```

How can a developer access the fullName property for dog?

A. `dog.fullName`

B. `dog.fullName()`

C. `dog.get.fullName`

D. `dog.function.fullName()`

147. A developer has a fizzbuzz function that, when passed in a number, returns the following:

- 'fizz' if the number is divisible by 3.
- 'buzz' if the number is divisible by 5.
- 'fizzbuzz' if the number is divisible by both 3 and 5.
- empty string if the number is divisible by neither 3 or 5.

Which two test cases properly test scenarios for the fizzbuzz function?

Choose 2 answers

A.

```javascript
let res = fizzbuzz(3);
console.assert(res === 'buzz');
```

B.

```javascript
let res = fizzbuzz(15);
console.assert(res === 'fizzbuzz');
```

C.

```javascript
let res = fizzbuzz(NaN);
console.assert(isNaN(res));
```

D.

```javascript
let res = fizzbuzz(Infinity);
console.assert(res === '');
```

148. A developer writes the code below to return a message to a user attempting to register a new username. If the username is available, a variable named msg is declared and assigned a value on line 03.

```javascript
function getAvailabilityMessage(item) {
  if (getAvailability(item)) {
    var msg = "Username available";
    return msg;
  }
}
```

What is the value of msg when `getAvailabilityMessage('newUserName')` is called and the username is *not* available?

A. "msg is not defined"

B. "newUserName"

C. "Username available"

D. `undefined`

149. Refer to the code below:

```javascript
let strNumber = '12345';
```

Which code snippet shows a correct way to convert this string to an integer?

A. let numberValue = Integer(strNumber);`

B. ` let numberValue = textvalue.cornteger(););`

C. `let numberValue = Number(textVelue);

D. ` let numberValue = (stunber) textveluer

150. Refer to the following array:

```javascript
let arr = [1, 2, 3, 4, 5];
```

Which two lines of code result in a second array, `arr2`, created such that `arr2` is a reference to `arr`?

Choose 2 answers

A. `let arr2 = arr.slice(0, 5);

B. `let arr2 = Array.from(arrl);`

C. `let arr2 = arr;`

D. `let arr2 = arrl.sort():

151. Refer to the code below:

```
01 funetion Person() (
02 thie. fixsttane = "Zoha'y
os}
05 Perscn.prototype = [
06 jek: x => ‘Developer’
onde
09 const myFather = new Peracn():
10 const result = myFather.firstiame + ' * + myFather.jeb()
```

What is the value of result after line 10 executes?

A. " John Developer"

B. " John undefined"

C. Error: myFather.job is not a function

D. "undefined Developer"

152. A developer is setting up a new Node.js server with a client library that is built using events and callbacks.

The library:

- Will establish a web socket connection and handle receipt of messages to the server

- Will be imported with `require`, and made available with a variable called `ws`.

The developer also wants to add error logging if a connection fails.

Given this information, which code segment shows the correct way to set up a client with two events that listen at execution time?

A.

```javascript
ws.connect(() => {
  console.log('Connected to client');
}).catch((error) => {
  console.log('ERROR', error);
});
```

B.

```javascript
ws.on('connect', () => {
  console.log('Connected to client');
});
ws.on('error', (error) => {
  console.log('ERROR', error);
});
```

C.

```javascript
ws.on('connect', () => {
  console.log('Connected to client');
});
ws.on('error', (error) => {
  console.log('ERROR', error);
});
```

D.

```javascript
try {
  ws.connect(() => {
    console.log('Connected to client');
  });
} catch(error) {
  console.log('ERROR', error);
}
```

153. Considering type coercion, what does the following expression evaluate to?

```javascript
true + 3 + '100' + null
```

```

A. "4100null"

B. 104

C. "4100"

D. "2200null"

154. A developer is trying to convince management that their team will benefit from using Node.js for a backend server that they are going to create. The server will be a web server that handles API requests from a website that the team has already built using HTML, CSS, and JavaScript.

Which three benefits of Node.js can the developer use to persuade their manager?

Choose 3 answers

A. Executes server-side JavaScript code to avoid learning a new language.

B. Performs a static analysis on code before execution to look for runtime errors.

C. Uses non-blocking functionality for performant request handling.

D. Ensures stability with one major release every few years.

E. Installs with its own package manager to install and manage third-party libraries.

155. Refer to the following code block:

```javascript
01 class Student {
02   constructor(name) {
03     this._name = name;
04   }
05   displayGrade() {
06     console.log(`${this._name} got 70% on test.`);
07   }
08 }
09
10 class GraduateStudent extends Student {
11   constructor(name) {
12     super(name);
```
```

```javascript
    this._name = "Graduate Student " + name;
  }
  displayGrade() {
    console.log(`${this._name} got 100% on test.`);
  }
}
let student = new GraduateStudent("Jane");
student.displayGrade();
```

What is the console output?

- Better student Jackie got 708 on t
- Uncaught ReferenceError
- Better student Jackie got 100% on test.
- Jackie got 708 on test.

156. A developer implements a function that adds a few values.

```javascript
function sum(num1, num2, num3) {
  if (num3 === undefined) {
    num3 = 0;
  }
  return num1 + num2 + num3;
}
```

Which three options can the developer invoke for this function to get a return value of 10?

Choose 3 answers

A. ` sum (5)  (5);`

B. ` sum () (10);`

C. `sum (5, 5,) ();`

D. `sum (10, ();`

E. sum () (5) (5);`

157. Refer to the code declarations below:

let st=1 = 'Java';

let st=2 = 'Script';

Which three expressions return the string `JavaScript`?

Choose 3 answers

A. '${str=1}${str=2}';

B. str=1.cmonat(str=2);

C. const({str=1, str=2});

D. str=1 + str=2;

E. str=1.join(str=2);